Human-Centred Mechanism Design for Scalable Oversight

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Abstract

As AI systems grow more capable and autonomous, *scalable oversight* becomes a central alignment challenge. Scalable oversight involves how humans can effectively supervise AI systems that exceed any individual's capacity to fully verify its behaviour. We focus on the approach of spot-checking and identify two dimensions along which it must scale: the volume of outputs requiring review and the *capability gaps* between what AI systems produce and what humans can readily evaluate. We find that effective spot-checking requires integrating perspectives from economic *mechanism design* and *human-computer interaction*. For example, the mechanism of selecting what deserves review shapes the human reviewer's cognitive context, and the reviewer's cognitive abilities should inform how instances for review are selected. Our analysis reveals a limitation of the bidirectional alignment framework: human adaptation to AI is not simply a matter of AI literacy, but is actively shaped by the design of the oversight process itself. We argue for designing alignment mechanisms that scaffold human cognition and improve system performance in tandem.

CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**.

Keywords

Human-AI alignment, scalable oversight

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1 Introduction

As AI systems grow more capable and autonomous, a central challenge is *scalable oversight*, or the problem of how humans can effectively supervise AI systems that operate at scales, speeds, and complexities that exceed any individual's capacity to fully verify [2, 33]. Scalable oversight is, at its core, an alignment problem: oversight is only as good as the human feedback that shapes and corrects AI behaviour, whether through explicit ratings and rankings [28, 30, 40], natural language evaluations [9, 31], or implicit behavioural signals [29]. This feedback needs to scale in two directions: in volume, to generate sufficient data to align large machine learning models while not overwhelming the humans in how much feedback is asked of them; and over capability gaps, to empower humans to oversee models that may be more capable than themselves on certain tasks.

Much effort has been expended on developing the algorithms and infrastructures required to train machine learning models on human feedback, from fitting reward models that serve as proxies of human judgements so that millions of judgements can be generated online during training without relying on expensive and time-intensive human raters [28, 36] to developing reinforcement and supervised learning algorithms to update model weights in the direction of those human preferences [12, 40]. Yet, these advances largely treat the human contributor as a fixed component—a source of labels whose cognitive processes, motivational states, and interaction contexts are outside the scope of the system design. We argue that this is a critical gap.

Human oversight is not a labelling task, but an iterative, dynamic process that requires active cognitive engagement, judgement under uncertainty, and accountability. But we lack systematic understanding of what conditions produce high-quality human feedback in this more complex reality. In other words, we still lack the understanding of how we might design the oversight process itself so that human judgements are as informative and reliable as possible. Two relevant bodies of knowledge bear on this question, yet have developed largely independently:

Mechanism Design: Economic theorists have long asked how to structure interactions between intelligent interacting agents with private information and their own interests, such that desirable outcomes for both parties emerge, a field known as mechanism design [26, 27]. In the scalable oversight context, the human and the model are engaged in a repeated strategic interaction: the model produces outputs, the human provides feedback, and this feedback shapes future model behaviour. The key question is how to design this "mechanism" so that both parties are incentivised to act in ways that track human social and moral norms. Too often in scalable oversight research is the focus on the AI models seeking to deceive or strategise maliciously in the oversight context. It is equally possible that the humans in this supervisory dynamic are ill-incentivised to provide high-quality, value-aligned feedback. We need mechanisms in which it is *optimal* for humans to give honest, high-quality judgements of model outputs, rather than satisficing or gaming the feedback process [19, 25].

Human-Computer Interaction: From HCI and cognitive science, we know that humans have cognitive limitations, emotional

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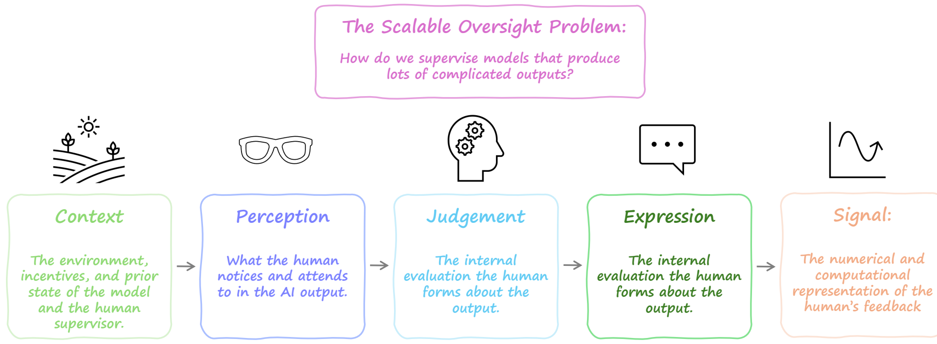

Figure 1: The scalable oversight problem has five phases: Context, Perception, Judgement, Expression, and Signal. We propose an integrated Human-Compatible Mechanism Design perspective that explores these components and their interactions.

needs, and contextual dependencies that shape their ability to perceive, understand, and communicate judgements [1, 38]. The design of the feedback environment (e.g., the interface, the task framing, the cognitive load imposed, the timing of requests) directly affects the quality of the feedback that a human can provide. HCI offers rich frameworks for understanding and designing for these factors, from theories of distributed cognition to empirical methods for evaluating human-AI collaboration [13, 22].

The Bidirectional Human-AI Alignment (BiAlignment) framework proposed by Shen et al. [33] identifies scalable oversight as one of three central challenges for alignment and emphasises the need for cross-disciplinary collaboration. However, the BiAlignment framework's treatment of the human side of alignment focuses primarily on AI literacy and cognitive adjustment—important goals, but ones that treat the human as an agent who must be *extrinsically educated* rather than a participant whose cognition can be *intrinsically scaffolded through good design*. In this paper, we argue that the combination of mechanism design and HCI provides a richer foundation for addressing scalable oversight (section 2). Then, we examine two concrete dimensions of scalable oversight—volume and capability gaps—and outline how mechanism design and HCI could address each (section 3). Finally, we reflect on what this implies for expanding the BiAlignment framework (section 4).

2 Scalable Oversight Problem Formulation

When a human overseer evaluates an AI system's output, several cognitive and communicative steps intervene between the AI's behaviour and the signal that the training or evaluation system ultimately receives. To identify where and how the quality of human oversight can be improved, we distill the oversight process into five stages (Figure 1):

(1) **Context.** The environment, incentives, and prior states that the human(s) and the AI system bring to the interaction (e.g., compensation schemes for annotators, accountability structures for deployed systems).
(2) **Perception.** What the human notices and attends to in the AI output. Given the volume and complexity of AI-generated content, selective attention is inevitable; the question is whether the human perceives the aspects of the output that are most relevant to the oversight task.
(3) **Judgement.** The internal evaluation the human forms about the output's quality, correctness, safety, or alignment with values. This is shaped by the human's mental model of the AI system [3], their domain expertise, cognitive biases [6], and the framing of the evaluation task.
(4) **Expression.** How the human communicates that judgement through the available interface, such as through a binary rating, a ranking, a natural language critique, or a more structured annotation. The expressiveness of the interface constrains which aspects of the human's judgement can be conveyed [33, 37].
(5) **Signal.** What the downstream system actually receives and can use. This is the numerical and computational representation of the human's feedback as consumed by a reward model, training algorithm, or evaluation pipeline. [12, 28, 30, 31, 40].

The quality of the human feedback can degrade at any of these stages, and with the wrong incentives, the AI systems may learn to exploit those degradations via reward-hacking or goal misgeneralisation [2, 21, 32]. Different disciplines have focused on different parts of this pipeline. Most alignment research has concentrated on the **Signal** stage: given a set of human preference labels, how should we train reward models and update the policy of the AI system [12, 28, 30, 31, 40]? This work generally takes the upstream stages as given. By contrast, mechanism design and HCI offer complementary lenses on the earlier stages where signal quality is actually determined.

Mechanism design would typically be focused on **Context** and **Expression**. The emphasis is on structuring the rules, incentives, and information flows of the interaction so that truthful, effortful participation is each party's best strategy, while defining the "message space" through which agents report their private information [26, 27]. In the context of scalable oversight, mechanism design asks: given that the human overseer has private knowledge (their **Judgement** of the AI's output) and their own interests (minimising effort while maximising compensation), how do we design the oversight game *qua* **Context** so that honest, high-quality **Expression** of those judgements is incentive-compatible?

HCI complements mechanism design. The role of the human from **Perception** to **Judgement** to **Expression** is akin to bridging the Gulf of Execution [14] in guiding an AI system. The human brings with them their mental models, informed by the **Context**. Along these lines, HCI asks how to design interactions, information architectures, and interfaces to shape what humans notice and attend to when providing feedback [1]. The **Signal** step is primarily focused on the incorporation of the human feedback into AI system behaviour. An open-ended question for scalable oversight is how to design mechanisms that help humans bridge the Gulf of Evaluation, i.e., to what extent the AI system incorporated the feedback.

Neither approach alone adequately addresses the full pipeline. Mechanism design's standard assumptions of rational agents, stable preferences, and unlimited cognitive capacity, break down at the **Perception** and **Judgement** stages, given the limitations of human cognition. HCI, meanwhile, may not prioritize explicitly modeling or conceptualizing the game theoretic aspects of scalable oversight, risking the potential for designing interactions that do not address real-world needs. Producing reliably high-quality human feedback for scalable oversight *requires* designing across the pipeline, combining mechanism design's attention to strategic context with HCI's attention to cognitive context, and grounding both in an understanding of human judgement under uncertainty. In the following section, we examine two concrete dimensions of scalable oversight through this integrated lens and show how entangled mechanism design and HCI considerations are.

3 Two Open Challenges with Scalable Oversight

Scalable oversight is challenging along two dimensions that, while conceptually distinct, interact in practice: (i) the sheer *volume* of AI-generated information that may require human review, and (ii) the potential *capability gaps* between what AI systems are producing and what humans can readily reason about. Consider a coding agent that the user has tasked with developing a new compiler [8]. Such an agent may write over 100,000 lines of code, which is impossible for the user to comprehensively review. Much of this code may even be beyond the user's knowledge or capabilities.

When comprehensive review is infeasible, AI system design and development rely on *spot-checking*: the practice of directing human attention to and validating select instances of an AI model's process or outcomes. Below, we examine the challenges associated with volume and capability gaps and sketch how an integrated human-centred mechanism design perspective could address them.

3.1 Volume

When AI systems generate outputs at scale, whether as steps in a long autonomous reasoning chain or as actions taken by an agent, comprehensive human review becomes infeasible. The volume problem crops up across the oversight pipeline. At the **Perception** stage, information overload leads to attentional narrowing: humans default to superficial heuristics, anchoring on the most salient features of an output rather than the most diagnostic ones [1]. At the **Judgement** stage, cognitive fatigue and test-related stress accumulate over extended review sessions, degrading the quality of evaluations over time, as observed in content-moderation [20], medical image decision-making [35], and judicial review [11]. At the **Expression** stage, time pressure to review more outputs incentivises coarse feedback over richer more effortful signals [39].

Volume makes the *selection* aspect of spot-checking critical. From a mechanism design perspective, the selection process must be robust to strategic behaviour. If the AI system has any influence over or foreknowledge of which outputs will be spot-checked, it may learn to concentrate quality on reviewed instances and cut corners elsewhere—a form of specification gaming, but applied to the oversight process [2, 21, 32]. Selection mechanisms therefore need to balance targeted review, in which instances are sampled based on the AI's uncertainty or through automated check flags, with unpredictability (e.g., random sampling) to maintain compatibility. These considerations start to resemble HCI research on crowdsourcing and assigning tasks to specific workers. How might we formalize these insights and apply them to the problem of scalable oversight?

Furthermore, volume makes the *inspection* aspect of spot-checking a critical HCI question. How should scalable oversight systems support the human in navigating the space of selected instances effectively? Schneiderman's mantra of "Overview first, zoom and filter, then details-on-demand" [34] is a well-studied paradigm that could help users detect patterns across large datasets of instances by leveraging human visual perception. An essential contribution is to design interfaces that help humans to rapidly triage AI outputs before investing cognitive effort into deep review.

An integrated human-centred mechanism design approach would jointly design the selection mechanism and the review interface. For example, the system might present the human with a curated set of instances chosen via a combination of uncertainty sampling and random auditing, displayed in an overview visualisation that highlights clusters of similar outputs and flags statistical anomalies. These instances can be screened to assign them to reviewers with the right knowledge and experience to judge them, trading off against recruitment costs. The humans' review pattern (which instances they choose to examine more closely, how much time is spent on them, what feedback they provide), then becomes an oversight signal and an input to the selection mechanism's future behaviour. This creates a feedback loop between the mechanism (which structures what is presented) and the interface (which structures how it is perceived and judged), one that can be designed to improve over time as the system learns which kinds of instances most benefit from human review. In this way, we can design human-compatible mechanisms that overcome the problem of volume while maintaining feedback quality and minimising the risk of exploitation by the AI system.

3.2 Capability gaps

The second dimension of scalable oversight concerns situations where the AI system's outputs are not just vast but also beyond the human reviewers ability to evaluate directly, either through lack of knowledge or the sheer complexity of the outputs. This raises the concern that, as AI systems become more capable, the gap between what they can do and what any individual human can verify widens [2, 4, 7]. Spot-checking requires directing human attention to components of the outputs that they are capable of judging.

Capability gaps create failures primarily at the **Judgement** stage. The human may perceive the AI's output perfectly well but lack the expertise or cognitive capacity to assess whether the output is actually correct, safe, or value-aligned. This has downstream consequences for **Expression** and **Signal**. Consider the case where a coding agent writes a function that uses an obscure but exploitable function from an out-of-date library. If the human is forced to make a decision about the safety of this function, then in the best case scenario they will answer randomly. In the worst case, they will make a systematic error because they are unaware that the referenced function is in fact exploitable, having used it when importing it from the updated library. The way that humans are able to express their judgements (e.g., with or without uncertainty estimations) modulates these risks, but all ultimately effect the accuracy of the **Signal**.

This is qualitatively different from volume: it cannot be solved by showing the human fewer instances or giving them better visualisations of the same information. It requires changing *what* information the human sees or changing *how* the evaluation task is structured so that it falls with in the human's competence.

Capability gaps are of particular concern if AI systems may cause danger or harm (e.g., create bioweapons). The AI safety community has proposed several mechanism-design-inspired approaches to resolve capability gaps. In one of the most promising, *debate* [5, 15, 17, 18], two copies of an AI model debate a claim, with one copy decomposing the claim into simpler sub-claims and the other evaluating the truth of those sub-claims. This process repeats recursively. A human judge then evaluates sub-components of the debate tree, which may involve thousands of branching rounds of debate. The key insight is that even if the human cannot verify a complex claim directly, they may be able to judge which of two competing arguments is more compelling, especially if the debate format incentivises each AI to expose weaknesses in the other's reasoning. Similar strategies have been proposed in recursive reward modelling [23] and iterated amplification [10].

Researchers have developed these approaches almost entirely from an algorithmic perspective, with little attention to the human factors that ultimately determine whether they work in practice. Debate, for instance, assumes that the human judge can follow sub-components of a potentially technical argument between two sophisticated AI systems. However, HCI research suggests that this needs complex scaffolding. Liu et al. [24] document how end-user programmers require natural language overlays in order to comprehend complex code. Jiang et al. [16] propose novel interfaces that allow participants to more easily parse complex language model outputs. What would the equivalent design interventions look like for an AI debate transcript? How should shared assumptions be surfaced, points of agreement highlighted, and the crux of disagreement made salient?

These tools point toward a general strategy for addressing capability gaps: rather than asking the human to evaluate outputs they cannot fully understand, *restructure the oversight task* so that it operates at a level of abstraction where human judgement is reliable, enabling a kind of spot-checking. This might mean asking the human to evaluate whether the AI's reasoning follows stated principles rather than whether its conclusions are correct; or to identify which aspects of an output they are and are not confident about, so the system can route uncertain aspects to more qualified reviewers or alternative verification methods.

An integrated human-centred mechanism design approach to capability gaps would jointly design (a) the decomposition or restructuring of the evaluation task (i.e., the mechanism design question of how to break a hard verification problem into pieces that are both human-tractable and incentive compatible), (b) the interface through which decomposed tasks are presented (i.e., the HCI question of how to scaffold comprehension, surface relevant context, and support appropriate confidence calibration [1, 33]), and (c) the aggregation of human judgements back into an oversight signal (i.e., the engineering question of how to combine human inputs into a reliable training or evaluation signal). Current approaches tend to design these components in isolation, with an attendant loss in generality and efficacy.

Scalable oversight is a problem that pulls in two directions: volume and capability gaps. As AI systems become more capable, they tend to produce more information (longer reasoning chains, more complex outputs), so capability gaps and volume pressures compound. Addressing scalable oversight in the long run will likely require not just better time management for spot-checking, but fundamentally new oversight paradigms, that treat the human not as a bottleneck to be minimised, but as a cognitive agent whose strengths (contextual judgement, value reasoning, anomaly detection) can be amplified through throughtful mechanism and interface design, while their limitations (bounded attention, domain-specific expertise, susceptibility to cognitive biases) are compensated for by the very same mechanisms.

4 Conclusion

We explore how mechanism design and HCI provide complementary perspectives on how to address two challenges core to scalable oversight: volume and capability gaps. We hone in on spot-checking as a form of oversight. A common pattern emerges: the mechanism should consider the cognitive context of the human reviewer, and the human's cognitive responses should inform and shape the mechanism. To date, the research concerning scalable oversight has largely taken up and designed the mechanistic and human-centric aspects in isolation. We argue that we need to take a more integrative stance.

This integrative perspective also exposes a limitation of the bidirectional alignment framework [33]. A core component of that framework is that humans align to AIs through strategies that largely reduce to AI literacy: understanding model capabilities and limitations in order to calibrate expectations. Our analysis of spot-checking suggests this is insufficient. For instance, when a scientist spot-checks an autonomous experimental campaign, or a developer audits a coding agent's output, their cognition is not merely being *informed*. Their cognition is being actively shaped by what the interaction entails and when it occurs. In other words, the human feedback loop shapes human cognition as much as human judgements shape the AI system's future behaviour.

We speculate that these insights extend beyond scalable oversight to alignment more broadly. We see an overlooked opportunity to design alignment mechanisms such that this entanglement is productive: where the process of interaction simultaneously scaffolds human cognition and improves system performance over time, rather than treating the human as a static judge to be educated and the AI as a system to be corrected. Both are evolving together.

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